

MALEK GHABI

Software & Data Developer | Apprentice at NOVOUTILS

Cluses, France | +33 6 04 10 21 56 | malekghabi.education@gmail.com

GitHub | **LinkedIn** | **Portfolio**

PROFILE

I hold a Bachelor's Degree in Computer Science from Avignon University and I am pursuing a Master's Degree in Computer Science at Université Catholique de Lille, specializing in Applied Computer Science and Intelligent Systems, through an apprenticeship at NOVOUTILS. I am interested in software development, data, databases and AI, with the goal of designing reliable, useful solutions aligned with business needs.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, PHP, C/C++, Bash, HTML5, CSS3
Back-End & APIs: Spring Boot, FastAPI, Flask, Laravel, REST APIs, JWT, WebSockets, Swagger/OpenAPI
Front-End: React, Vite, Tailwind CSS, Bootstrap, AJAX, Chart.js
Databases: PostgreSQL, MySQL, SQLite, Supabase, Firebase, data modeling and migration
Data & automation: Pandas, NumPy, Scikit-Learn, Excel, Power BI, file processing, data visualization
DevOps & tools: Git, GitHub, Docker, Docker Compose, GitHub Actions, Linux, Nginx, Postman, VS Code, IntelliJ
Methods: OOP, MVC architecture, UML, Agile/Scrum, testing, technical documentation
Soft skills: Autonomy, rigor, organization, curiosity, teamwork, adaptability

TECHNICAL PROJECTS

Aura Student Life OS — React • Spring Boot • PostgreSQL • Docker

Full-stack SaaS platform for organizing student life: finances, goals, planning, habits, sport tracking and an AI assistant.

- Full-stack architecture: React/Vite frontend, Spring Boot REST API (Swagger/OpenAPI), PostgreSQL and Docker.
- Spring Security/JWT authentication, interactive dashboards, Excel/PDF exports and an AI assistant connected to the API.
- Modular design bringing several business domains together in a global dashboard of personal indicators.

Snake AI — 24h to Code Hackathon — JavaScript • Canvas • A* • Q-Learning

Advanced version of the Snake game built in 24 hours, featuring an adaptive artificial intelligence.

- Evolving AI Beast whose behavior adapts to the player through reinforcement learning (Q-Learning).
- Optimal pathfinding with A* and smart points fleeing the Snake using the Boids algorithm.
- Combined several AI algorithms in real time without performance loss, with a smooth Canvas interface.

EasyMonitor — Bash • Python • Flask • SQLite • Chart.js

Distributed system monitoring tool tracking multiple virtual machines through a web interface.

- Bash/Python agent collecting CPU, RAM, disk, network and user metrics, sent to a Flask server over secured HTTP.
- SQLite/RRDTool storage, real-time and historical charts, and automatic alerts (CPU, memory, full disk).
- Network communication between two VMs, agent authentication and optimized collection frequency.

WithYou — React • Laravel • Socket.io • Supabase

Collaborative real-time synchronized video-watching platform, built by a team of 5 — MVP in progress.

- Synchronized video playback (YouTube IFrame API), virtual rooms and real-time chat with Socket.io and Laravel Reverb.
- Laravel API with JWT authentication, role-based member permissions, polls and public room listing.
- 3-service architecture (React/Vite client, Laravel API, real-time service), deployment planned on Netlify/Render/Railway.

Search Engine — Java • Indexing • TF-IDF

Simplified search engine with corpus analysis, indexing and relevance ranking.

- Text normalization (case, punctuation, stop-words) and inverted index construction (term → documents).
- TF-IDF weighting, cosine similarity and top-K result ranking on Wikipedia and book corpora.
- Two vocabulary modes (full or reduced) and an API to easily plug in new corpus sources.

PROFESSIONAL EXPERIENCE

Software & Data Engineer Apprentice — NOVOUTILS, Thyez — September 2026 – Present

- Redesigning a database architecture connected to the CEGID PMI ERP: analysis, structuring and data migration.
- Enhancing a Power BI sales-tracking solution: dashboards, business KPIs and automated reporting.
- Designing a measurement equipment management application: access control, logging, data backup and notifications.
- Contributing to the coordination of an AI-driven software project, aligning business needs and technical specifications.

Full-Stack Developer Intern — CEA Marcoule — May 2026 – August 2026

- Developed CADI Web, an internal application modernizing INSTN/CEA training processes.
- Built React/TypeScript interfaces and Python/FastAPI APIs integrating existing processing tools.
- Automated Excel, Word and CSV file processing with progress tracking and application logging.
- Contributed to software architecture, testing, debugging and version control with Git.

EDUCATION

Master's Degree in Computer Science — Applied Computer Science and Intelligent Systems — 2026 – 2028
Université Catholique de Lille — apprenticeship at NOVOUTILS

Bachelor's Degree in Computer Science — 2023 – 2026
Avignon University, France

Bachelor of Science in Computer Science — 2025 – 2026
University of Geneva, Switzerland — dual enrollment alongside Avignon University, first year completed

French Baccalaureate — Science track — 2023
Jean-Henri Fabre High School, France

LANGUAGES

French: fluent | English: professional | Arabic: fluent | Italian: intermediate